

Hands on 2

Hibernate XML Configuration & Object-Relational Mapping (ORM)

Hibernate lets Java objects map to database tables. In XML configuration, this mapping is done using .hbm.xml files.

Example Mapping (Employee.hbm.xml):

xml

CopyEdit

```
<hibernate-mapping>

  <class name="com.example.Employee" table="EMPLOYEE">

    <id name="id" column="ID">
      <generator class="native"/>
    </id>

    <property name="firstName" column="FIRST_NAME"/>
    <property name="lastName" column="LAST_NAME"/>
    <property name="salary" column="SALARY"/>
  </class>
</hibernate-mapping>
```

- ◆ class name → Java class
- ◆ table → Database table
- ◆ property → Maps class field to column
- ◆ id → Maps primary key
- ◆ generator → Auto-generates ID (e.g., native, increment, identity, etc.)

Core Hibernate Components and APIs

1. SessionFactory

- It is a **heavyweight** object created once per application.
- Built using the configuration (hibernate.cfg.xml), it holds details about the DB connection and mappings.
- It is responsible for creating Session objects.

java

CopyEdit

```
SessionFactory factory = new Configuration()
```

```
.configure("hibernate.cfg.xml")
```

```
.addResource("Employee.hbm.xml")
```

```
.buildSessionFactory();
```

2. Session

- A **lightweight**, short-lived object used to interact with the DB.
- Represents a single unit of work (1 transaction = 1 session typically).

java

CopyEdit

```
Session session = factory.openSession();
```

3. Transaction

- Hibernate wraps DB operations in transactions.
 - You **must begin a transaction** even for SELECT in some databases.
-

4. beginTransaction()

- Starts a DB transaction.

java

CopyEdit

```
Transaction tx = session.beginTransaction();
```

5. commit()

- Commits the transaction (saves changes to DB permanently).

java

CopyEdit

```
tx.commit();
```

6. rollback()

- Reverts the changes if an exception occurs.

java

CopyEdit

```
tx.rollback();
```

7. session.save(Object)

- Persists a new Java object to the database table.
- Assigns the generated ID to the object.

java

CopyEdit

```
Employee emp = new Employee("Lavanya", "Y", 25000);  
session.save(emp);
```

8. session.createQuery().list()

- Used to run HQL (Hibernate Query Language) queries.

java

CopyEdit

```
List<Employee> list = session.createQuery("FROM Employee").list();
```

 FROM Employee → Fetches all rows from the Employee table (mapped via Employee.hbm.xml)

9. session.get(Class, id)

- Fetches a single object by its **primary key**.

java

CopyEdit

```
Employee emp = session.get(Employee.class, 101);
```

10. session.delete(Object)

- Deletes the given object from the DB.

java

CopyEdit

```
Employee emp = session.get(Employee.class, 101);  
session.delete(emp);
```